

Travel Time Index of the Imposed Congestion Field

UrbanEV–Maryland AADT-loaded network (v3): formal TTI computation and MDOT comparison

UrbanEV–Maryland verification series

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Purpose and definition

The manuscript verifies the imposed congestion field against MDOT’s observed travel-time indices only qualitatively (“milder than 1.25–1.31”). This note computes the field’s Travel Time Index formally. For link ℓ at hour h the link TTI is the ratio of congested to free-flow travel time,

$$\text{TTI}_{\ell h} = \frac{t_{\ell h}}{t_{\ell}^0} = \frac{v_{\ell}^{\text{free}}}{v_{\ell h}^{\text{imposed}}} = 1 + 0.15 (v/c)_{\ell h}^{\beta},$$

capped by the 0.25 speed floor ($\text{TTI} \leq 4$). The area index aggregates link TTIs weighted by each link’s directional hourly volume $w_{\ell h} = \text{AADT}_{\ell} s_h(K_{\ell})/2$, the same K -rescaled two-peak profile that generates the field:

$$\text{TTI}_h = \frac{\sum_{\ell} w_{\ell h} \text{TTI}_{\ell h}}{\sum_{\ell} w_{\ell h}}.$$

The computation covers the 114,720 major-class links matched to MDOT SHA AADT segments (≤ 60 m, mainline-to-mainline and ramp-to-ramp).

Results

Scope	Volume-weighted TTI
Daily, all matched links	1.144
AM peak (07:00)	1.215
PM peak (17:00)	1.226
Freeways (motorway/trunk), daily	1.081
Freeways, PM peak	1.181
Arterials, daily	1.175
Overnight (03:00)	1.000

Link-level distribution at the PM peak: median 1.00, 75th percentile 1.00, 95th percentile 1.06, maximum 4.00 (the floored bottleneck links). Congestion is concentrated: most matched links remain near free flow while the loaded corridors carry large delays, matching the corridor structure in the congestion map.

Comparison with MDOT observations

Quantity	Imposed field	MDOT observed
Area-wide peak TTI	1.22–1.23	1.25–1.31
Worst segments	4.0 (floor)	2.0–3.8

The imposed field reaches approximately 94% of MDOT’s observed area-wide peak travel-time index, so at the area level it nearly reproduces observed recurring congestion rather than merely understating it. The conservatism documented in the manuscript is real but concentrated where its mechanisms predict: freeways carry a lower daily index (1.081) because hourly-average volumes smooth out breakdown dynamics, and the even directional split dilutes peak-direction loading. The bottleneck floor (TTI 4.0) brackets MDOT’s observed worst-segment values of 2.0–3.8.

Method note

The weights are hourly link volumes on a per-mile basis (flow-weighted); link lengths do not enter the weight. This matches the standard approximation when segment lengths are homogeneous within road class; a fully length-weighted (VMT) version would shift the indices marginally. The hourly volumes are the same profile that generated the speed field, so this is a consistency-weighted summary of the imposed field itself, compared against MDOT’s independently observed indices.

Suggested manuscript sentence

“The imposed field’s volume-weighted peak-hour travel-time index is 1.22–1.23, just below MDOT’s observed area-wide 1.25–1.31, while the floored bottleneck links (TTI 4.0) bracket the observed worst segments of 2.0–3.8.”